

*MA.6.DP.1.2***Benchmark****MA.6.DP.1.2 Given a numerical data set within a real-world context, find and interpret mean, median, mode and range.**

Example: The data set {15, 0, 32, 24, 0, 17, 42, 0, 29, 120, 0, 20}, collected based on minutes spent on homework, has a mode of 0.

Benchmark Clarifications:

Clarification 1: Numerical data is limited to positive rational numbers.

Connecting Benchmarks/Horizontal Alignment

- MA.6.NSO.2.1, MA.6.NSO.2.2, MA.6.NSO.2.3
- MA.6.AR.2.2, MA.6.AR.2.3
- MA.7.DP.1.1, MA.7.DP.1.2

Terms from the K-12 Glossary

- Data
- Mean
- Median
- Mode
- Range

Vertical Alignment**Previous Benchmarks**

- MA.5.DP.1.2

Next Benchmarks

- MA.8.DP.1.1

Purpose and Instructional Strategies

In previous courses, students interpreted numerical data, with whole-number values, represented with tables or line plots by determining the mean, median, mode and range. In grade 6 accelerated, students find and interpret mean, median, mode and range given a numerical data set with positive rational numbers. Students also continue building their knowledge while using measures of center and measures of variability to compare two populations. In future courses, students are introduced to numerical bivariate data and depict it with line graphs and lines of fit.

- Instruction includes using numerical data generated by statistical questions.
- Instruction includes the understanding that data sets can contain many numerical values that can be summarized by one number such as mean, median, mode or range.
- Instruction includes data sets that have more than one mode. Students should understand that a mode may not be descriptive of the data set as a whole.
- Instruction includes how to calculate and interpret the mean (*MTR.4.1*).
- Instruction focuses on statistical thinking that allows for meaningful discussion of interpreting data (*MTR.4.1*). Students should be asked:
 - What do the numbers tell us about the data set? Is it representative of the whole?
 - What kinds of variability might need to be considered in interpreting this data?
 - What happens when you do not know all the measures in your data set?
 - Can you find missing data elements?

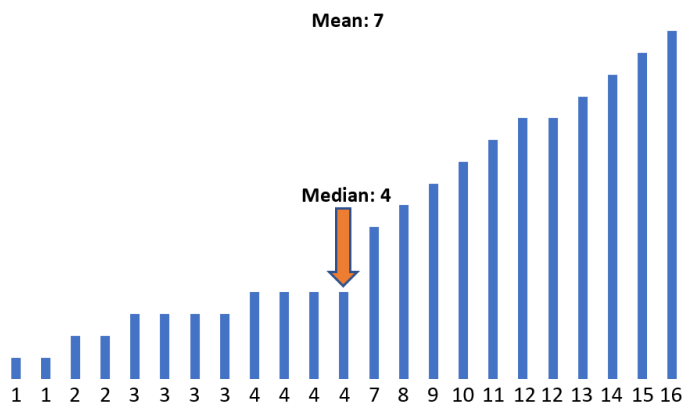
- Instruction includes student understanding of the difference between measures of center (mean and median) and measures of variability (range).
- Instruction includes students knowing when a number represents the spread or variability of the data, or when the number describes the center of the data.
- Instruction includes multiple representations of a given data set such as tables, lists, sets and graphical representations.
- Instruction includes graphical representations both horizontally and vertically and focus on box plots, histograms, stem-and-leaf plots and line plots.
- Instruction includes interpreting the median as splitting the data into two halves (50% above and below the median).

Common Misconceptions or Errors

- Students may incorrectly believe that “average” only represents the mean of a data set. Statisticians prefer to use the term “mean” rather than “average” since the term “average” is often used to refer to “typical” or “in the middle”.
- Students may confuse mean, median, mode and range.
- Students may neglect to order the numbers in the data set from least to greatest when finding the median or range.

Strategies to Support Tiered Instruction

- Teacher discusses with students how the use of the word average in daily life may show a different meaning of the word each time, just as other mathematical words have different meanings in everyday life. This will help students to understand that “average” does not only represent the mean of a data set.
- Teacher provides instruction focused on measures of center, co-creating anchor chart or graphic organizer.
- Teacher provides examples visually that show the clear middle of a data set, but where the mean is not the same. This visual will help students understand that the middle of a data set does not indicate that the amount is the mean.
 - For example, the figure below shows a data set with a mean of 7 and a median of 4.



Instructional Tasks

Instructional Task 1 (MTR.4.1, MTR.5.1)

Salena has 15 students in her class. The mean shoe size is 8.5. She records the shoe sizes below, with one missing:

7.5, 10, 15, 6.5, 7, 9, 9.5, 6, 11, 8.5, 7, 12, 6, 6.5, ____

What shoe size is missing? Explain how she can find the missing shoe size.

Instructional Task 2 (MTR.5.1)

Brandi is in the Girl Scouts, and they are selling cookies. There are 11 girls in her troop. The median number of boxes of cookies sold by a girl scout is 26, and the range of the number of boxes sold is 30 boxes.

Part A. What is a possible set of boxes of cookies sold?

Part B. If the mean number of boxes of cookies sold is 22, describe some possible characteristics of the data set.

Instructional Items

Instructional Item 1

Jonathan works for a sporting goods store. He is asked to report his sales of running shoes for each day of the week, in dollars. His numbers are given below.

{150.25, 122.85, 171.01, 118.48, 108.52, 130.15, 154.36}

Part A. What is the mean?

Part B. What is the median?

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*